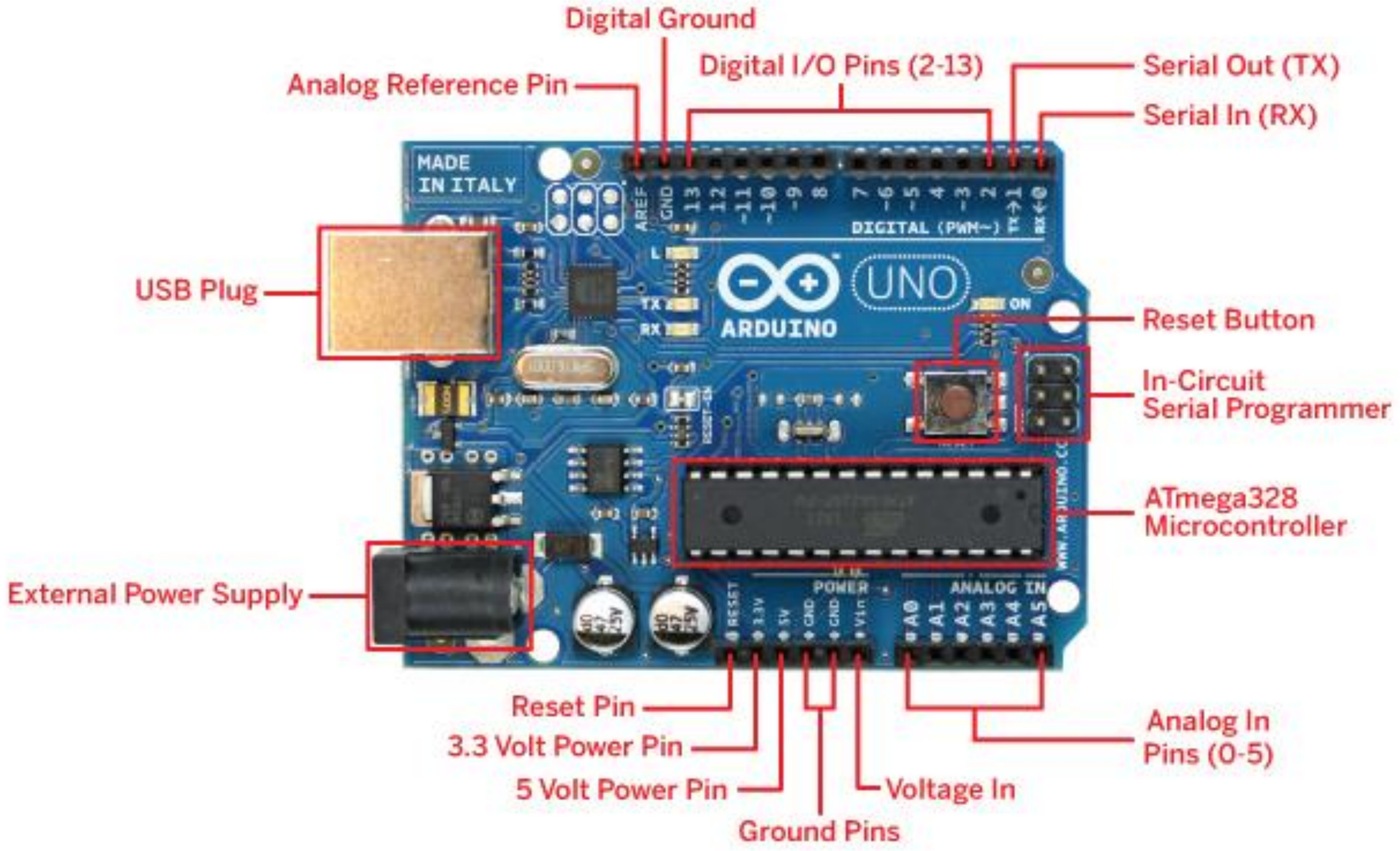
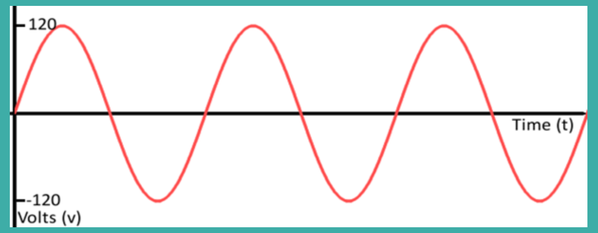
Arduino Analog & Sensing!

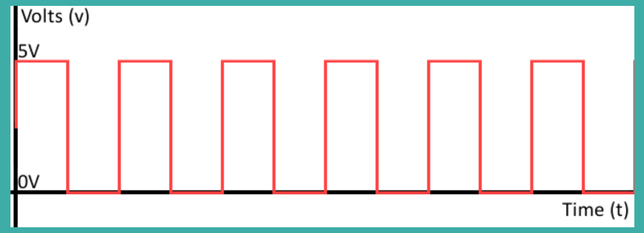
RECAP


Digital Ground
Analog Reference Pin
Digital I/O Pins (2-13)
Serial Out (TX)
Serial In (RX)
USB Plug
Reset Button
In-Circuit
Serial Programmer
ATmega328
Microcontroller
External Power Supply
Reset Pin
3.3 Volt Power Pin
5 Volt Power Pin
Ground Pins
Voltage In
Analog In
Pins (0-5)
MADE
IN ITALY
ARDUINO
UNO
DIGITAL (PWM~)
POWER
ANALOG IN

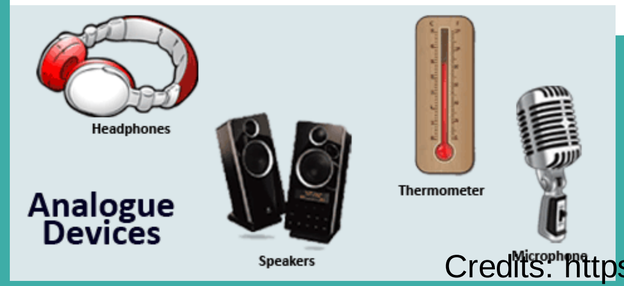
ANALOG



DIGITAL

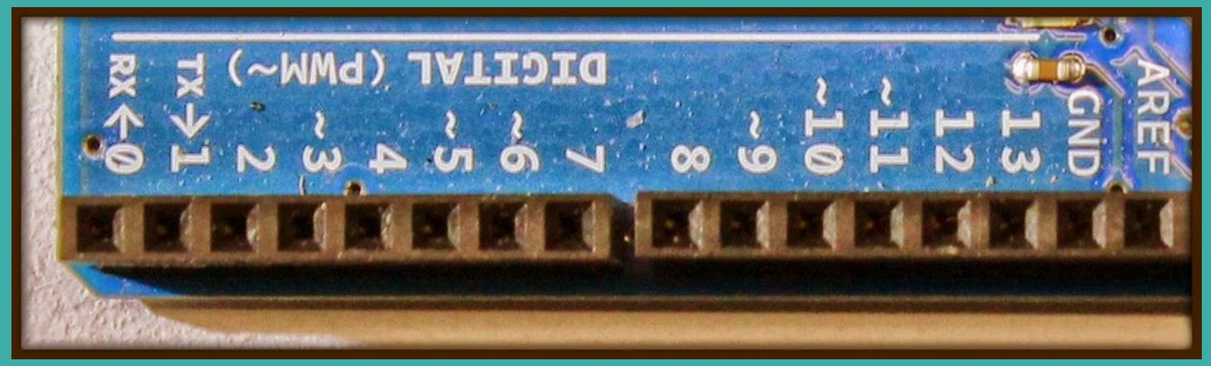


Signal	Analog signal is a continuous signal which represents physical measurements.	Digital signals are discrete time signals generated by digital modulation.
Waves	Denoted by sine waves	Denoted by square waves
Representation	Uses continuous range of values to represent information	Uses discrete or discontinuous values to represent information
Example	Human voice in air, analog electronic devices.	Computers, CDs, DVDs, and other digital electronic devices.
Technology	Analog technology records waveforms as they are.	Samples analog waveforms into a limited set of numbers and records them.



Credits: https://www.diffen.com/difference/Analog_vs_Digital

Digital pins



On board 14 Digital Pins are present

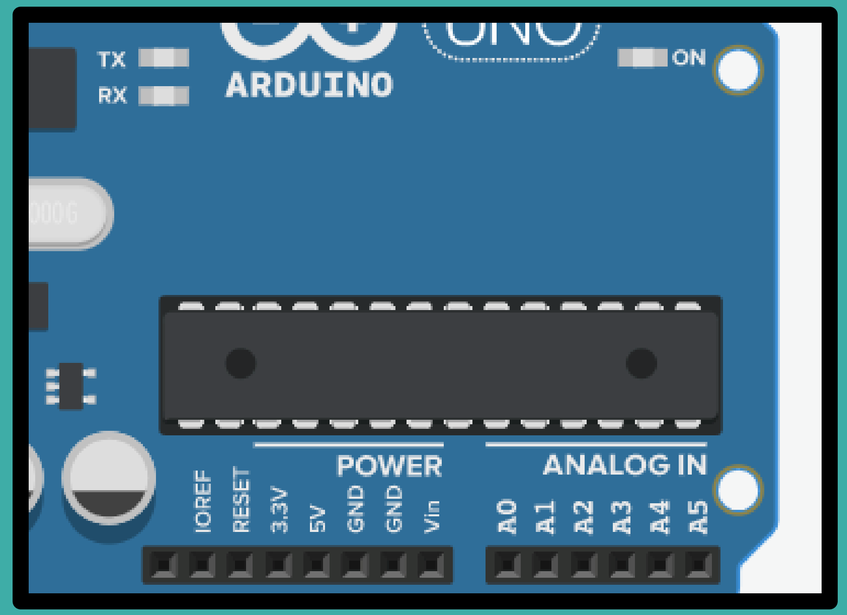
Generally, provides only two kinds of output i.e.

High = 5 Volt = 1

Low = 0 Volt = 0

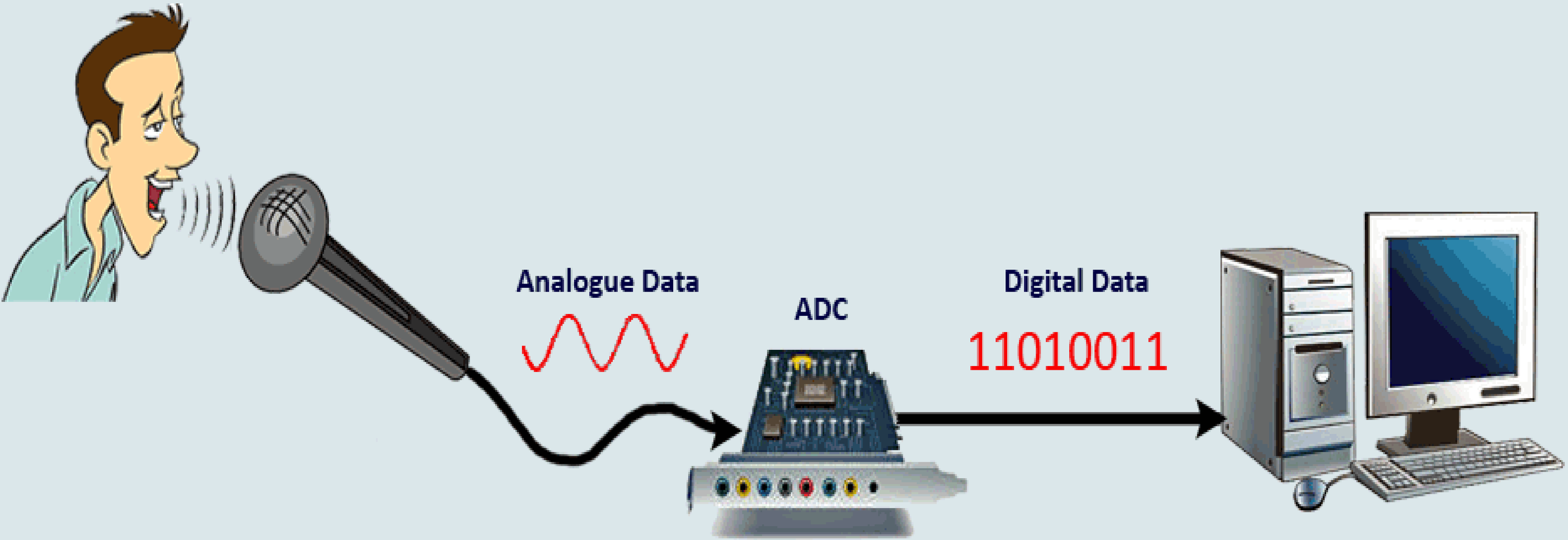
Onboard 6 channel analog-to-digital (A/D) converter.

ARef: On this pin, user defined reference value of ADC is given (Not more than 5 Volt).



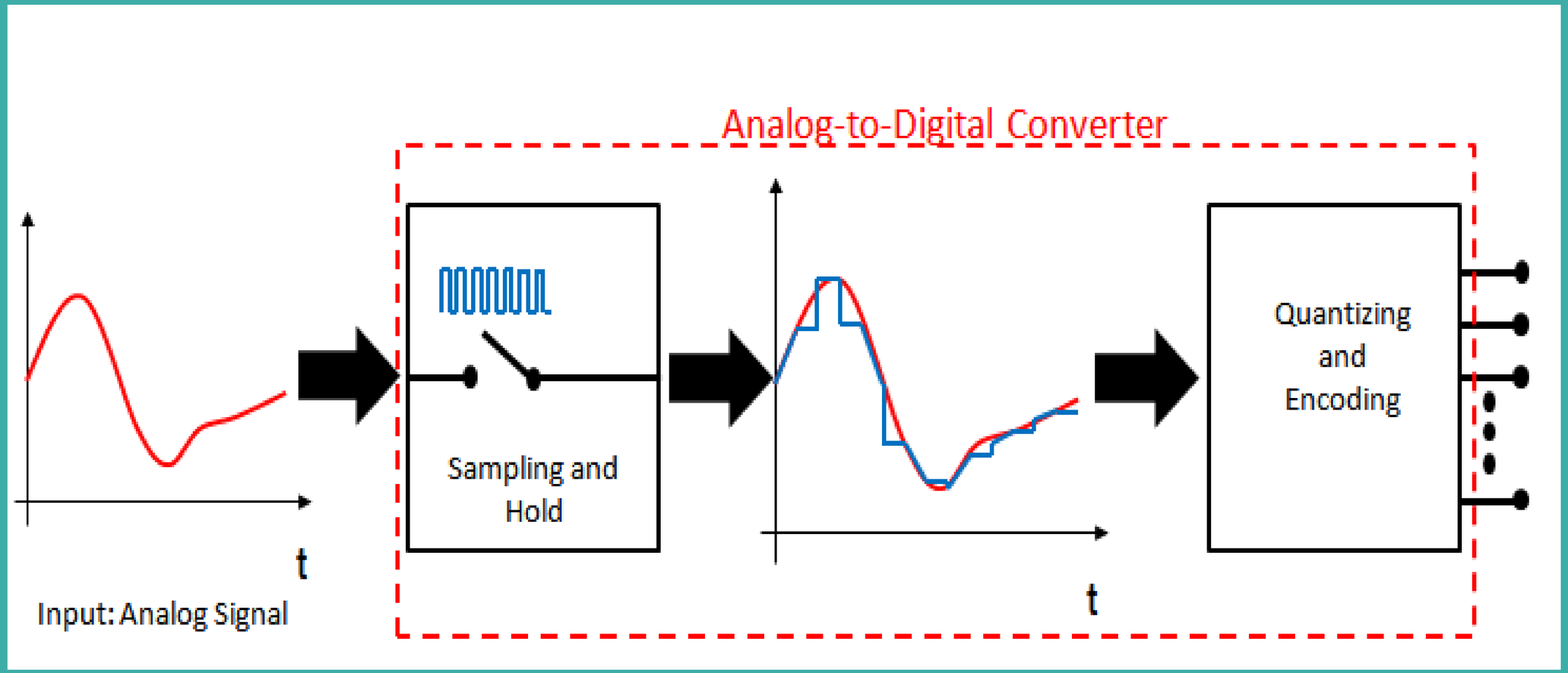
Analog pins

Analog to Digital Converter(ADC)



ADC process

Two main steps of the process
1.Sampling and Holding
2.Quantization and Encoding



Quantizing

The number of possible states that the converter can output is: $N=2^n$ (**n** is the number of **bits** in the AD converter)

Example: For a 3 bit A/D converter, $N=2^3=8$.
Analog quantization size:
$Q=(V_{max}- V_{min})/N = (10\ V - 0\ V)/8 = 1.25V$

Discrete Voltage	Ranges(V)
0	0.00-1.25
1	1.25-2.50
2	2.50-3.75
3	3.75-5.00
4	5.00-6.25
5	6.25-7.50
6	7.50-8.75
7	8.75-10.00

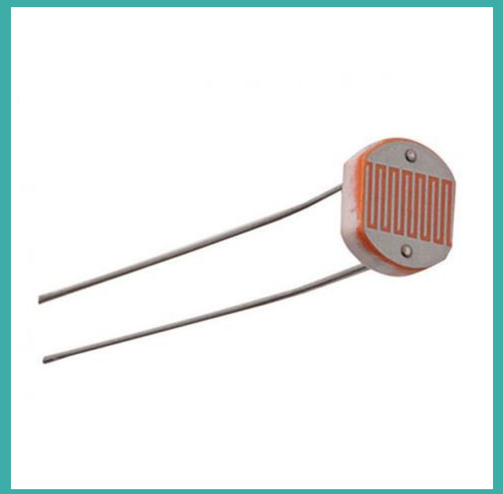
Encoding

Here we **assign the digital value** (binary number) to each state for the computer to read.

Output States	Output Binary Equivalent
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

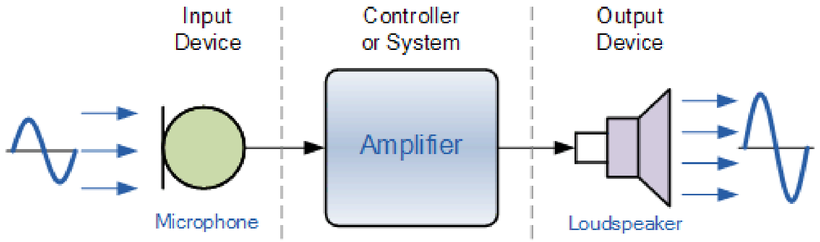
Sensing

Detects and responds to some type of input from the physical environment



Transduction

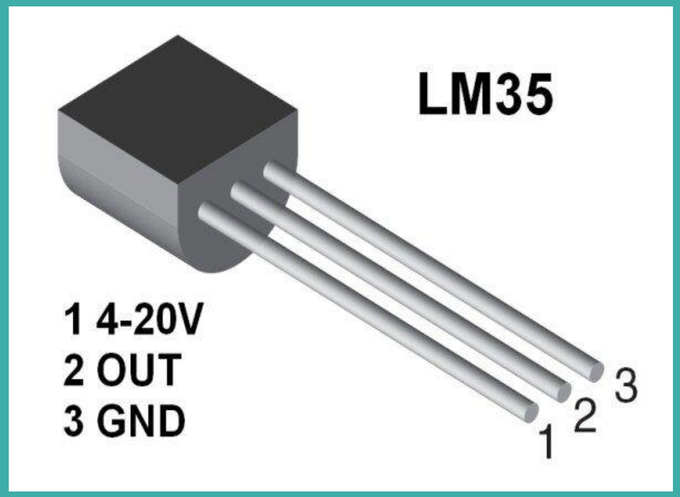
Converting one form of energy to another



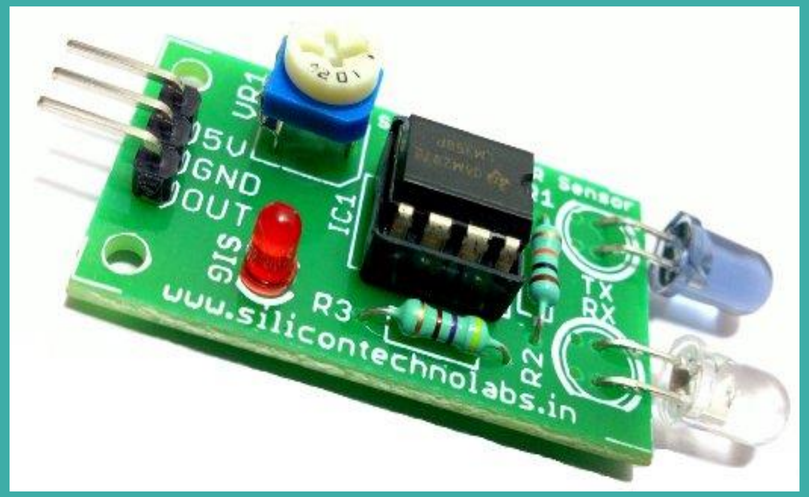
Actuation

To put into motion or action

Types of Sensors

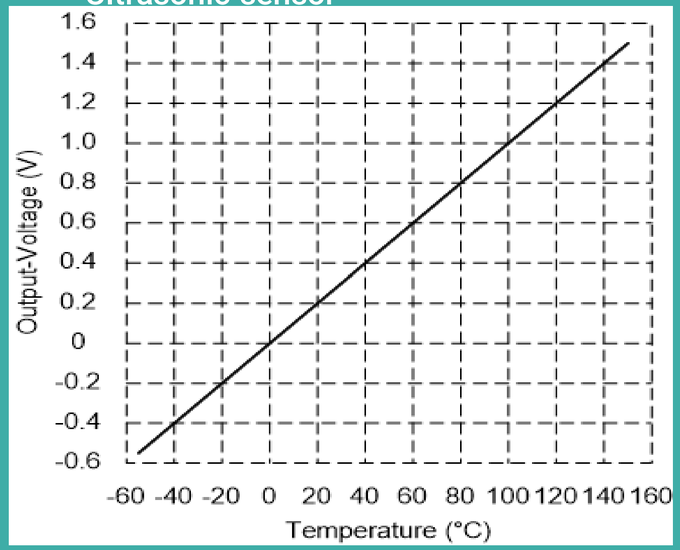


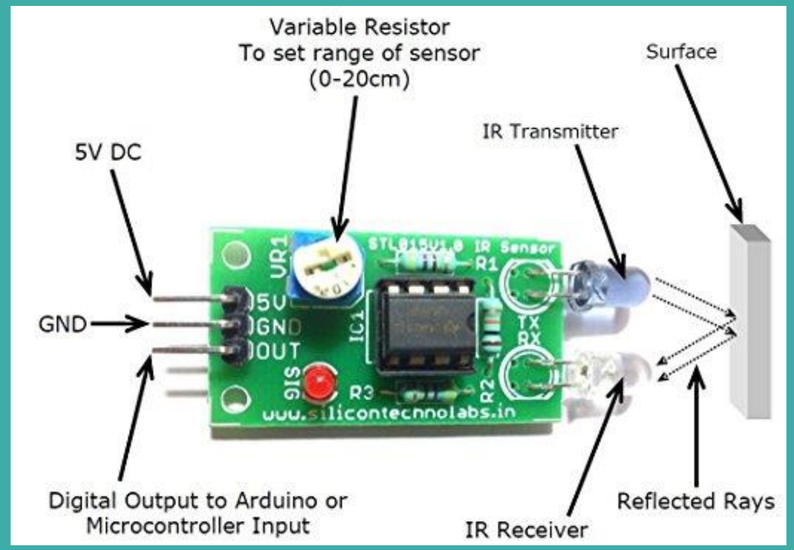
Temperature sensor(LM35)

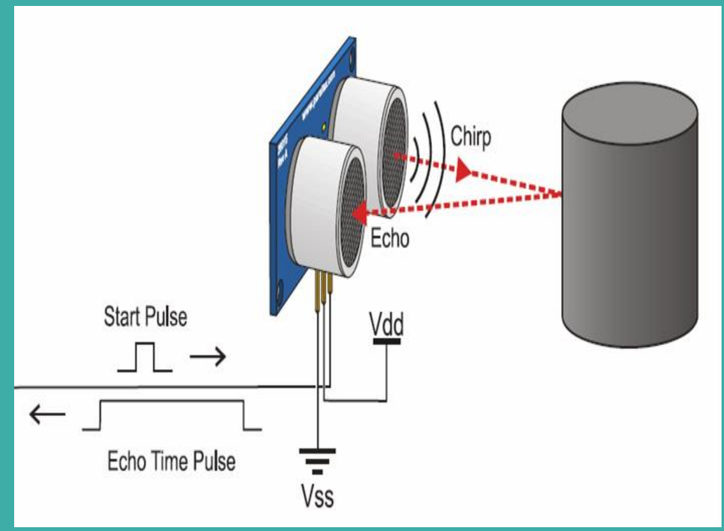


IR Sensor

Ultrasonic sensor







For more sensors click here

Count the sensors

Classifications of Sensing

Direct and hybrid transduction

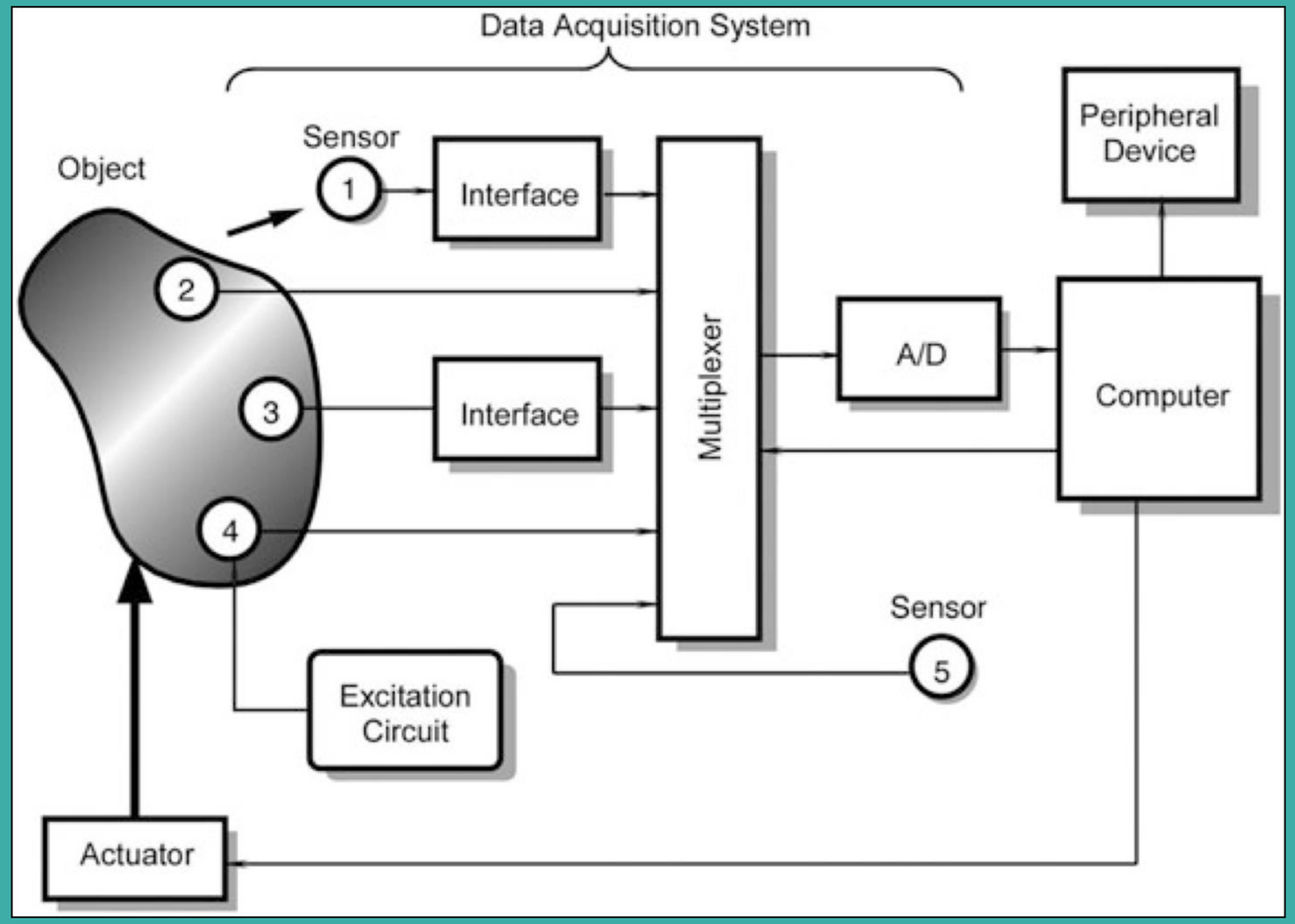
Absolute and relative

Active and passive

Intrinsic and extrinsic

Contact and noncontact

Data Acquisition System



http://www.kelm.ftn.uns.ac.rs/literatura/si/pdf/HandbookOfModernSensorsPhysicsDesignAndApplications.pdf

Analog Commands on Arduino

SENSING

- digitalRead() works on all pins. It will just round the analog value received and present it to you. If analogRead(A0) is greater than or equal to 512, digitalRead(A0)will be 1, else 0.

- analogRead() works only on analog pins. It can take any value between 0 and 1023.

ACTUATION

- digitalWrite() works on all pins, with allowed parameter 0 or 1. digitalWrite(A0,0) is the same as analogWrite(A0,0), and digitalWrite(A0,1) is the same as analogWrite(A0,255)

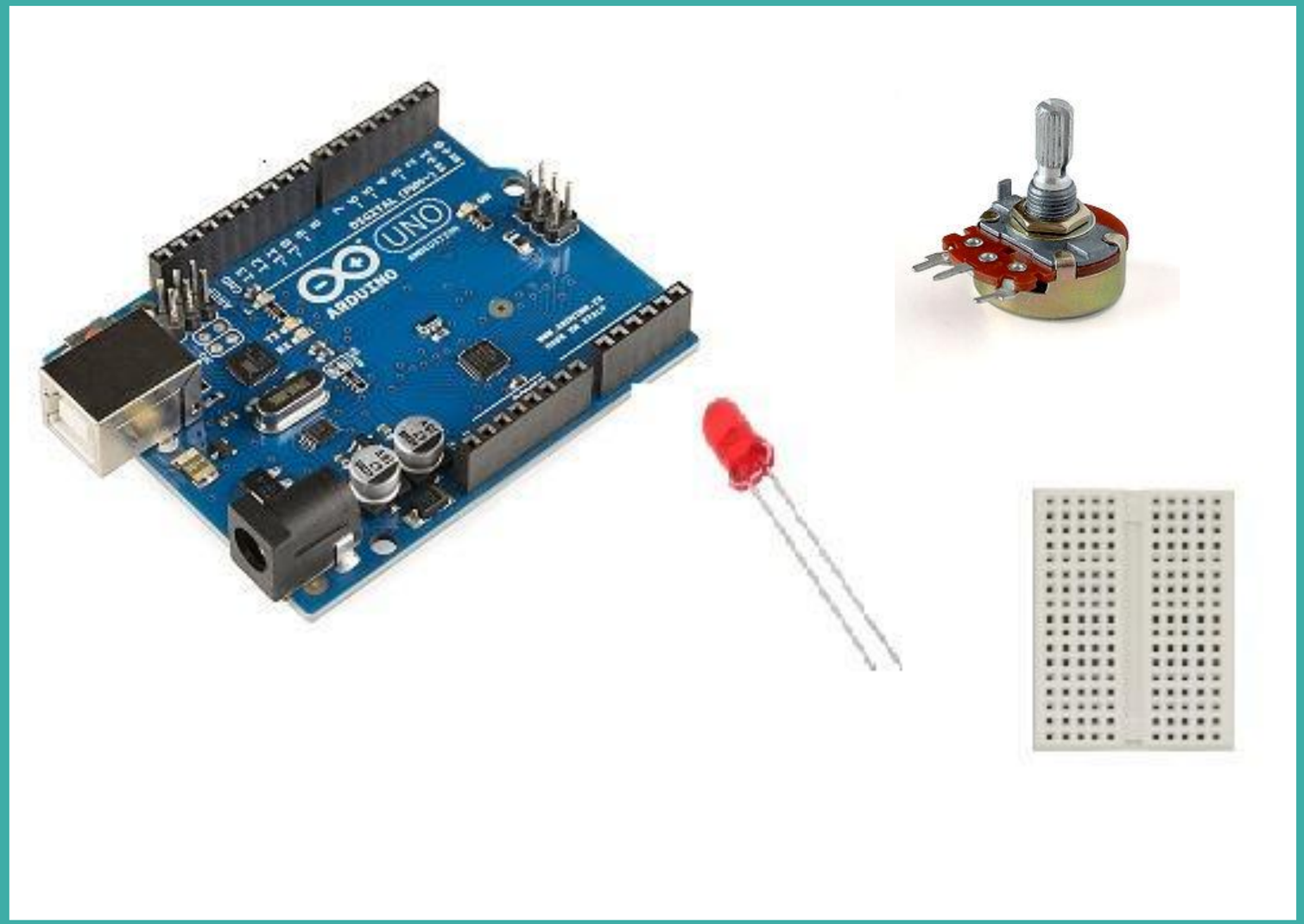
- analogWrite() works on all analog pins and all digital PWM pins. You can supply it any value between 0 and 255.

Activity-1

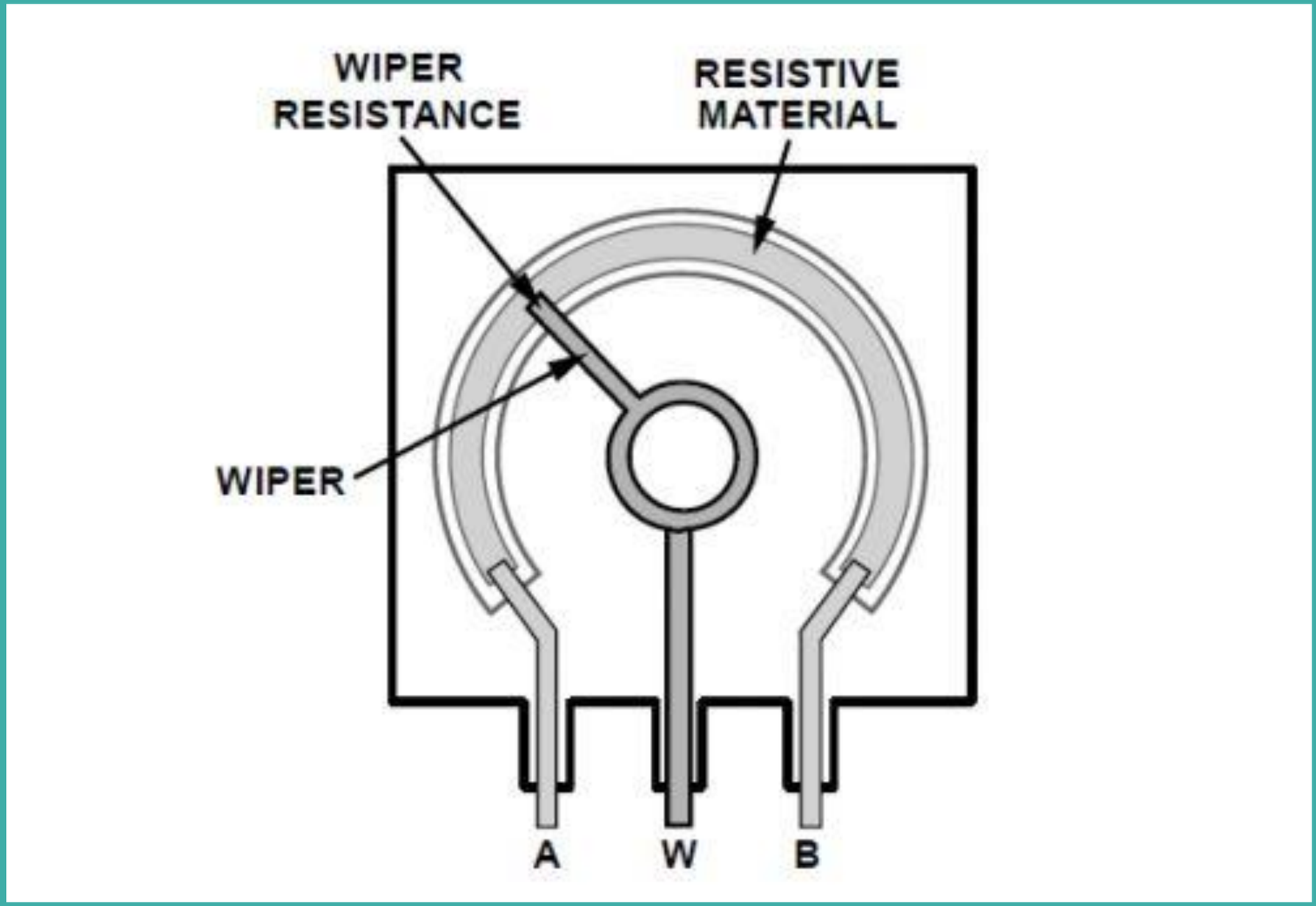
Control LED blinking using Potentiometer

Step 1: What You Will Need

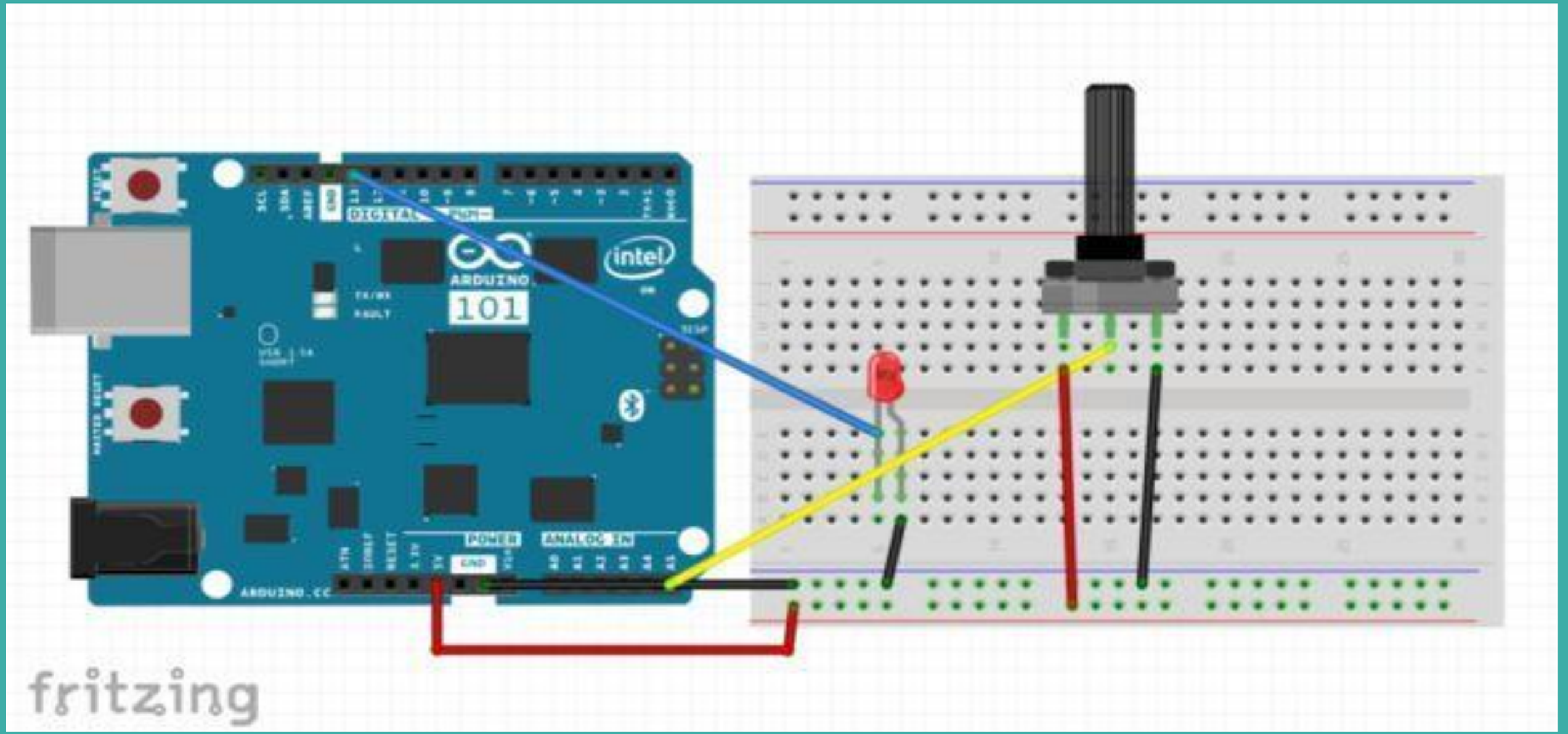
1.Arduino Uno
2.Potentiometer
3.Breadboard
4.LED
5.Connecting wires



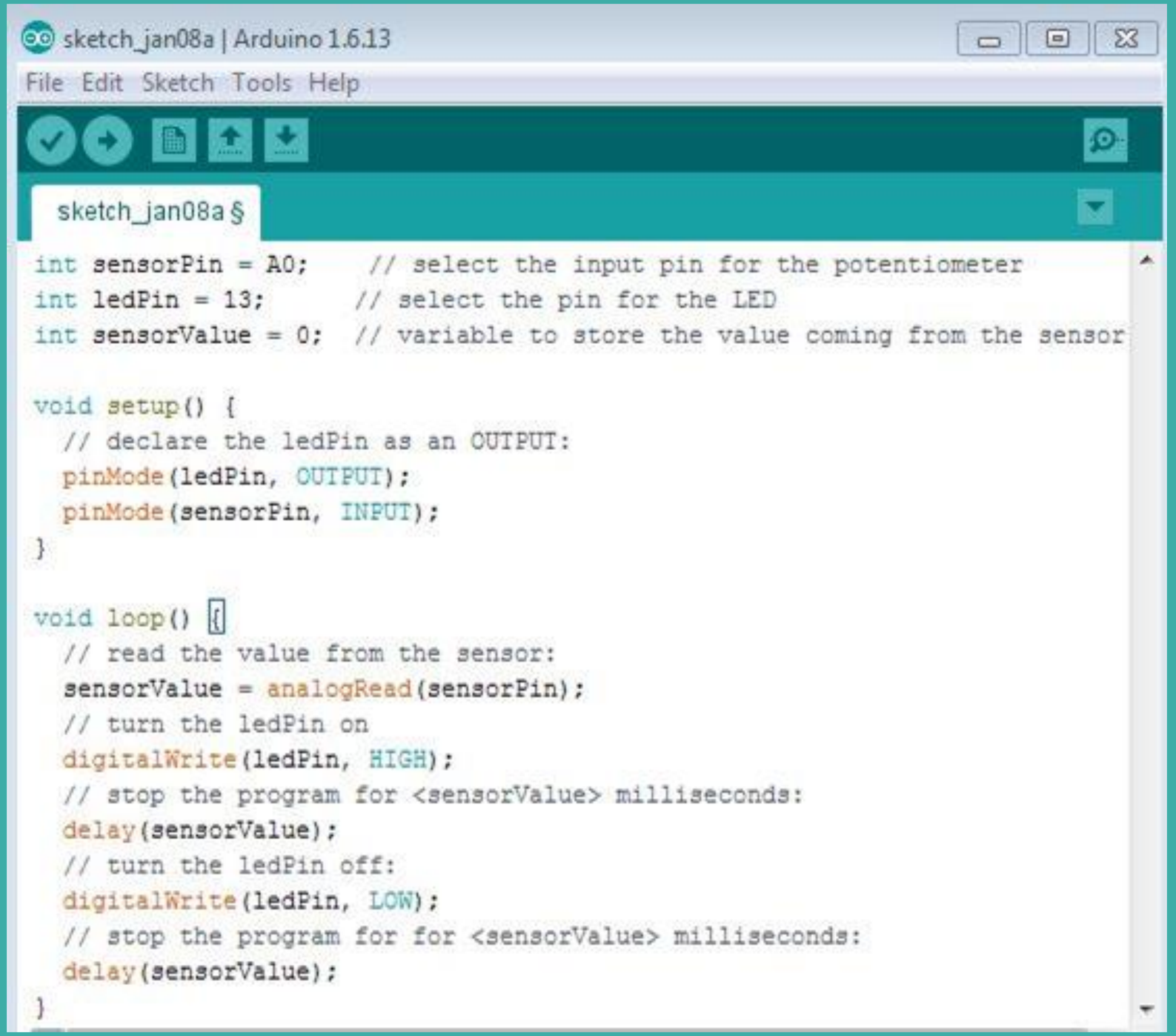
Potentiometer Black Box



Step 2: Circuit



Step 3: IDE Code

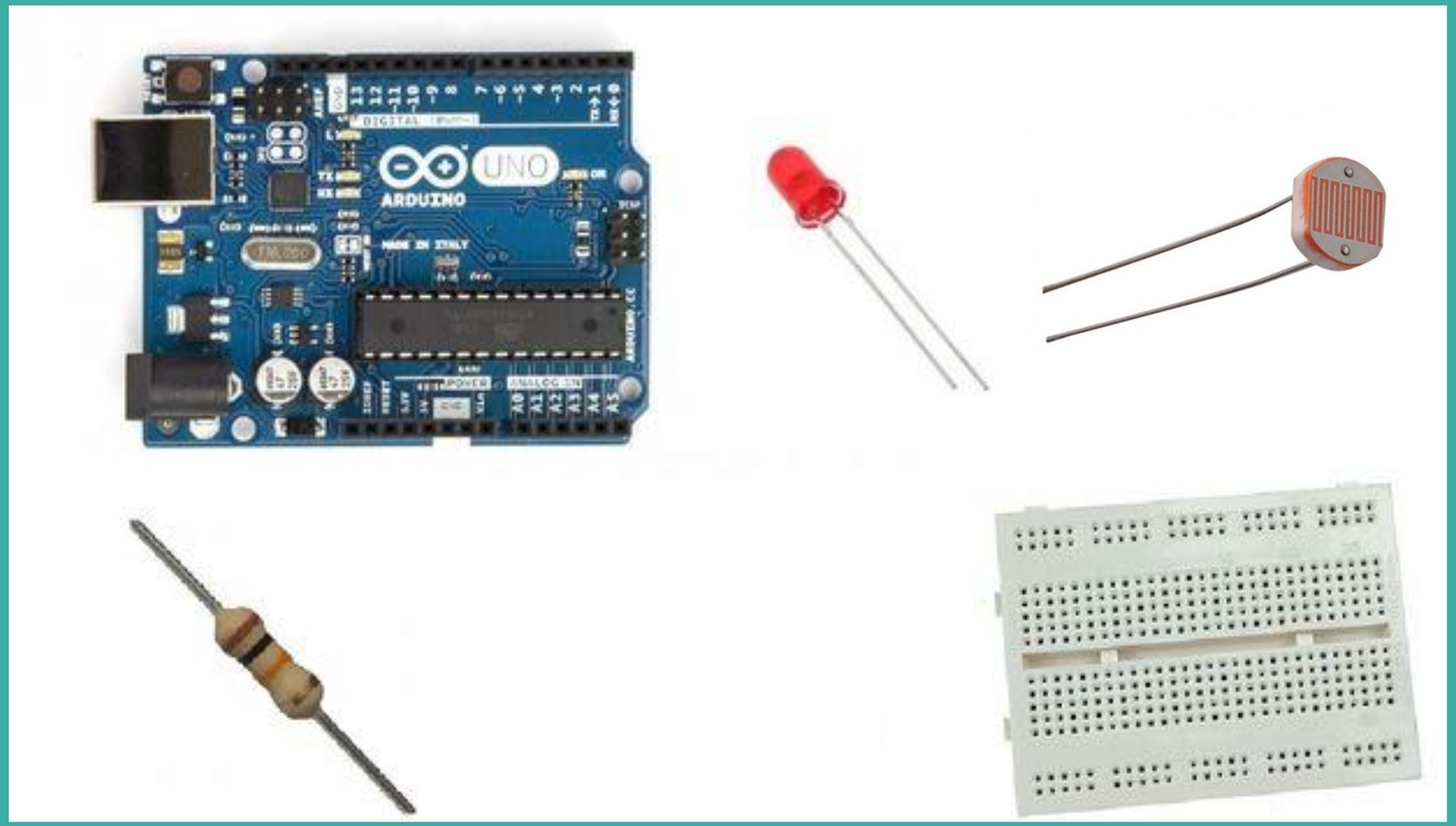


Activity-2

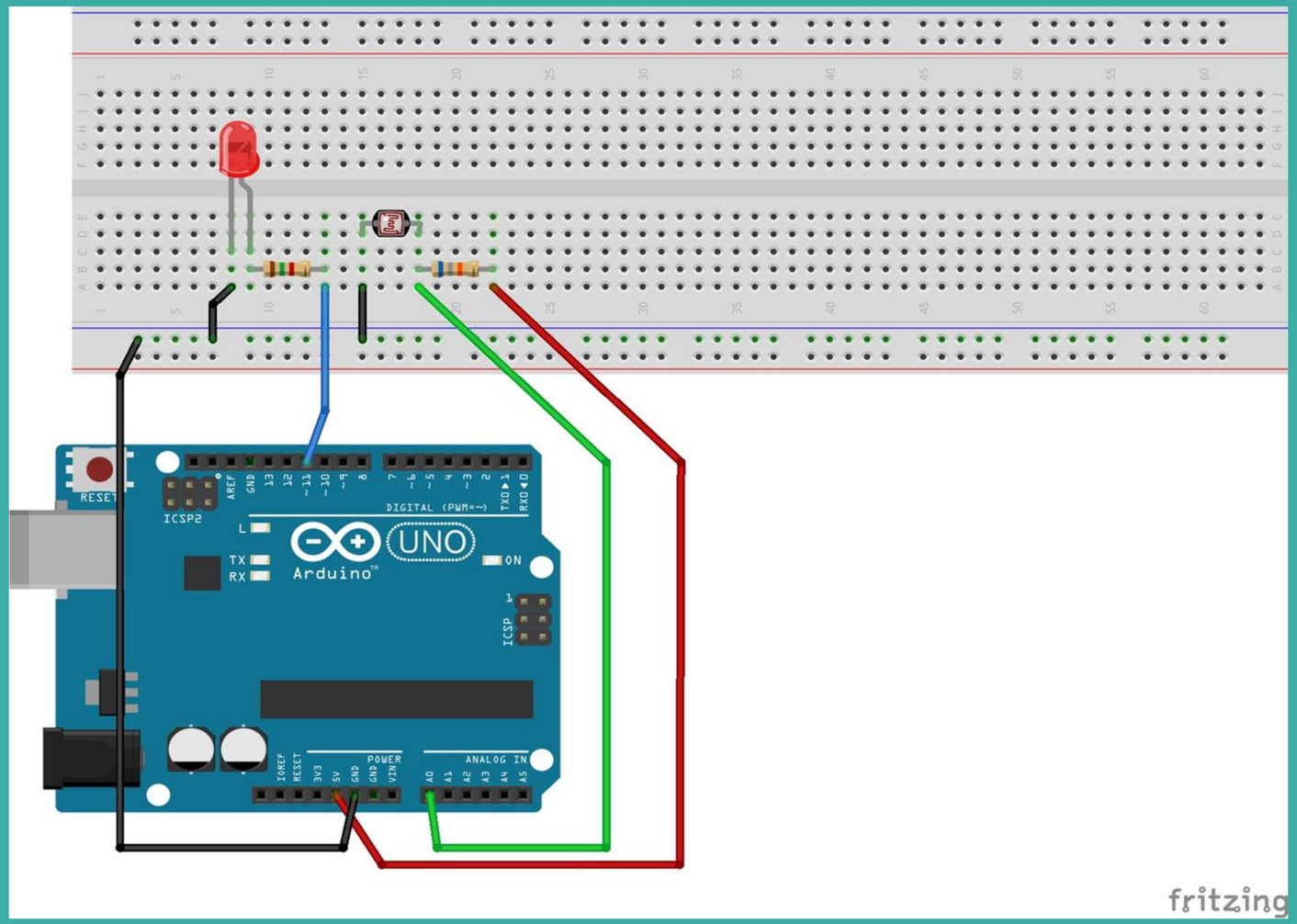
Control LED with a Light-Dependent Resistor (LDR)

Step 1: What You Will Need

1. An Arduino Uno
2. LDR
3. Breadboard
4. 10K resistor
5. Connecting wires
6. LED
7. 22O Ohm Resistor



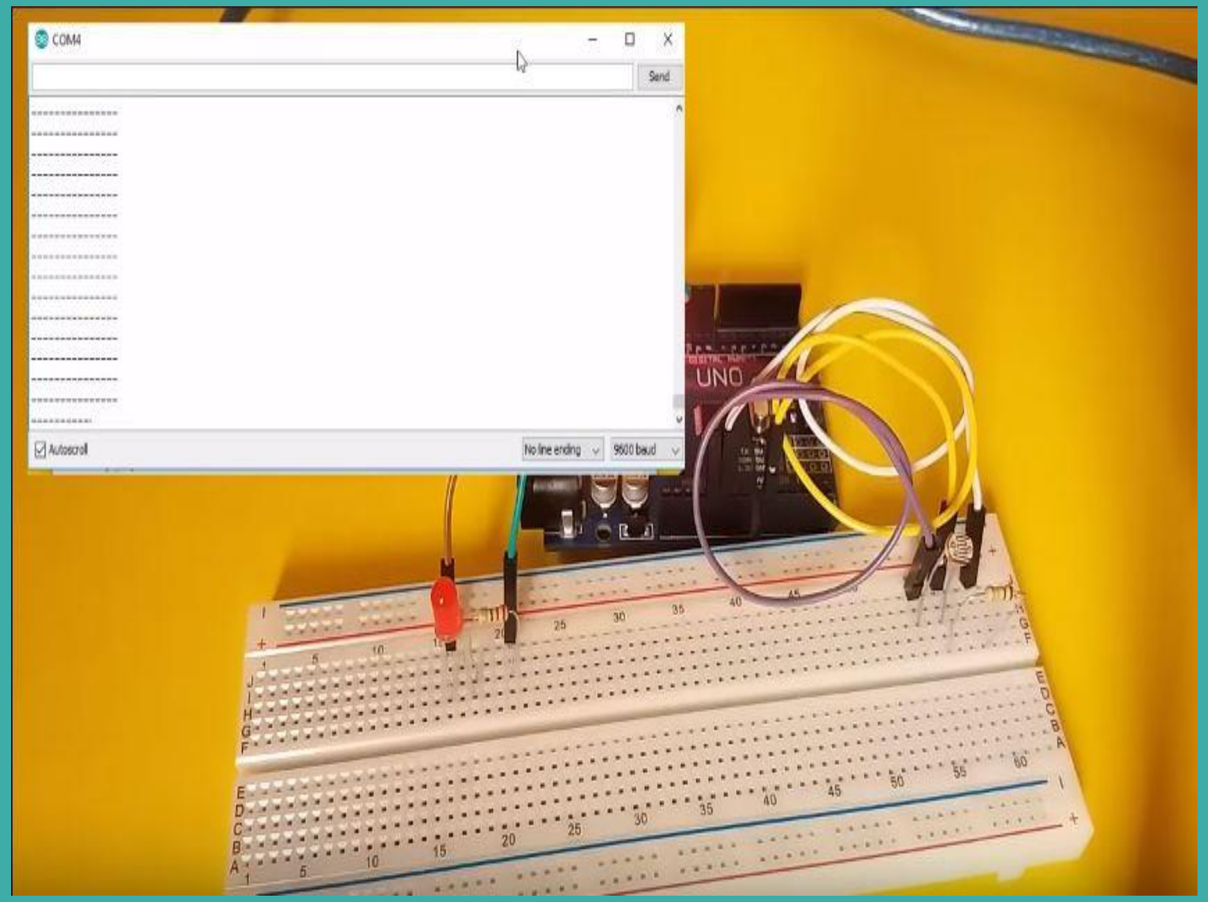
Step 2: Circuit

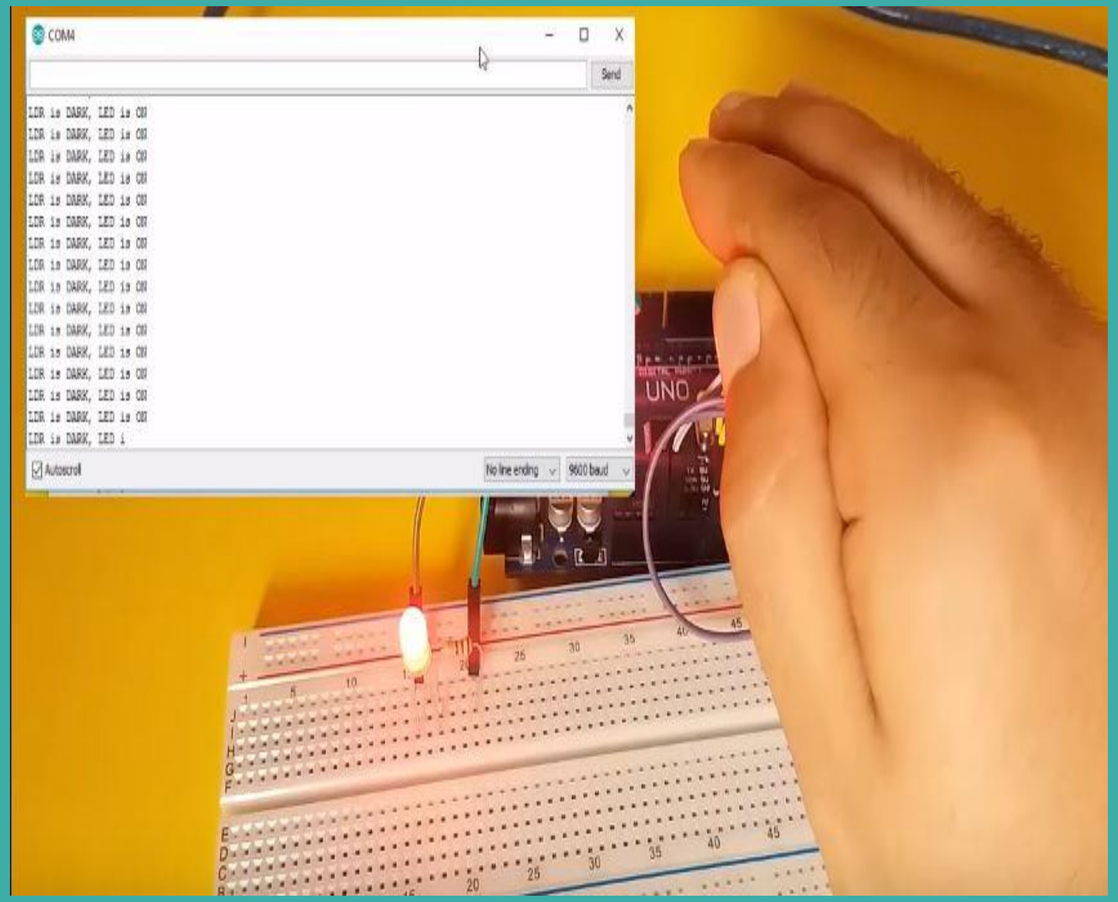


Step 3: IDE Code

```
const int ledPin = 13;
const int ldrPin = A0;
void setup() {
        Serial.begin(9600);
        pinMode(ledPin, OUTPUT);
        pinMode(ldrPin, INPUT);
}
void loop() {
    int ldrStatus = analogRead(ldrPin);
    if (ldrStatus <=300) {
        digitalWrite(ledPin, HIGH);
        Serial.println("LDR is DARK, LED is ON");
        delay (500);
    } else {
        digitalWrite(ledPin, LOW);
        Serial.println("---------------");
        delay (500);
    }
}
```

Expected Outcome





https://www.youtube.com/watch?v=4fN1aJMH9mM